

Ultrasonic Embossment of Acrylic Sheets with Transparency Control

Ayaka Tsutsui
University of Tsukuba
Tsukuba, Japan
ayakatsutsui1128@digitalnature
.slis.tsukuba.ac.jp

Tatsuki Fushimi
University of Tsukuba
Tsukuba, Japan
tfushimi@digitalnature
.slis.tsukuba.ac.jp

Kengo Tanaka
University of Tsukuba
Tsukuba, Japan
kengo.tanaka@digitalnature
.slis.tsukuba.ac.jp

Takahito Murakami
University of Tsukuba
Tsukuba, Japan
takahito@digitalnature
.slis.tsukuba.ac.jp

Yoichi Ochiai
University of Tsukuba,
R&D Center for Digital Nature
Tsukuba, Japan
wizard@slis.tsukuba.ac.jp

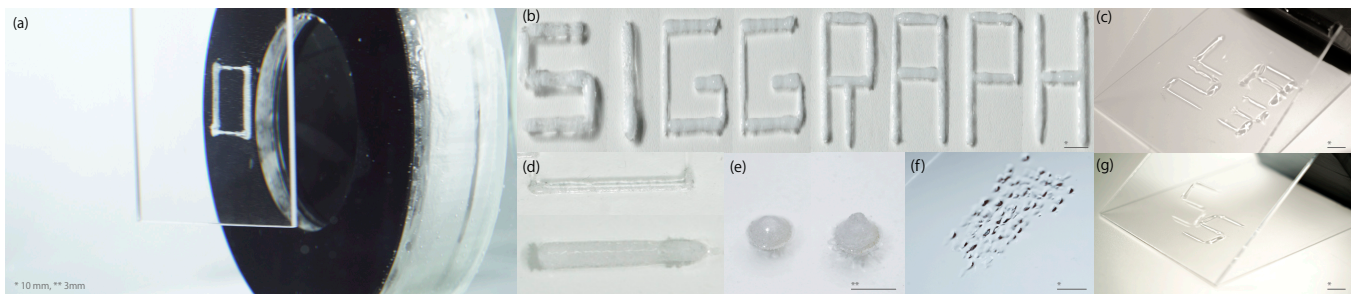


Figure 1: Examples of ultrasonic embossment generated by the system. (a) Creating a white square using HIFU. (b) Using XYZ plotter the letters outlining 'SIGGRAPH' was drawn by irradiating the acrylic sheet. (c) Demonstration of shadow art showing 'LA23'. (d) The thickness of the embossment can be adjusted by changing the distance between the plate and the transducer. (e) The embossment can either be round or sharp. (f) By randomizing the irradiation pattern, the acrylic surface can be textured. (g) Demonstration of shadow art showing 'S'. The transparency of the letter can be adjusted by changing the parameters.

ABSTRACT

This paper discusses a new processing technology for acrylic sheets using high-intensity focused ultrasound (HIFU). Acrylic are widely used due to their transparency and weather resistance, but current processing methods cannot create embossing features without reductive manufacturing methods. HIFU emits ultrasonic waves that melt the surface of the acrylic sheet, creating bumps that can be controlled by adjusting the irradiation time, speed, distance, and amplitude. This technology can control transparency, line thickness, and tactility and can create bumps and depressions instantaneously, making it potentially useful for embossing, etching, and braille generation. The paper demonstrates the controllable variables of the HIFU approach and highlights its potential for industry as a high-precision acrylic plate processing technology.

CCS CONCEPTS

• **Human-centered computing** → *Activity centered design*.

KEYWORDS

Ultrasonic, HIFU, Fablication, Manufacturing, Productivity

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1 INTRODUCTION

Acrylic is widely used in various fields such as architecture, design, and industry due to their excellent transparency and weather resilience. However, conventional processing methods are either reductive (laser cutting, CNC), additive 3D printers and cannot easily render minute embossing features (mm scale) with molds.

Herein, we propose to use high-intensity focused ultrasound (HIFU) as a new processing technology for acrylic sheets, and demonstrate embossment of acrylic sheets. Previous research using HIFU for processing has included 3D printing of structures

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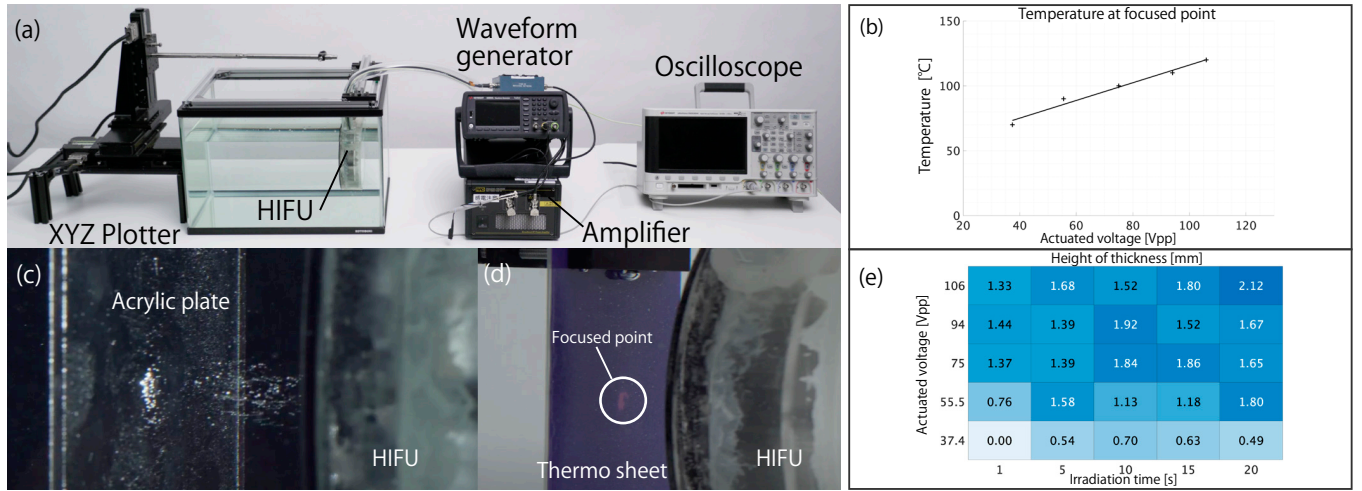


Figure 2: Experimental setup and the controllable parameters. (a) Device setup. 2MHz signal is generated by the waveform generator, and amplified by fixed gain amplifier. The T-junction is used to split the amplified signal to oscilloscope and HIFU. Acrylic sheet is attached to the XYZ plotter. (b) Shows the temperature at the focal point measured by the thermosheet. (c) Shows close-up picture of acrylic sheet when irradiated by focused ultrasound. A small cavitation is formed near the focal points. (d) Thermosheet was used to visually measure the temperature at the focal point. (e) The measured height of embossment depending on the irradiation time and actuated voltage in millimeters.

using acoustic cavitation [Habibi et al. 2022] and forming of 3D objects [Melde et al. 2023]. Differently, ultrasonic wave emitted by HIFU are used to melt the surface of the acrylic sheet with thermoviscous effects, and creates a bump on top of the sheet. This technology can easily control the transparency, line thickness, and tactility of acrylic sheets by changing the irradiation time, speed, distance of travel and amplitude of acoustic wave. It can instantaneously (<1 second) form bumps and depressions; making it suitable for processes such as embossing or etching. This is potentially useful for the generation of braille, and optical components, and is expected to contribute to industry as a high-precision acrylic plate processing technology.

2 METHOD

We conducted experiments in deionized water using HIFU transducer (Sonic Concept’s H-106). The experimental parameters included a fixed frequency of 2MHz, water temperature of 20°C and HIFU geometric focal length of 63.2mm. All acrylic board had thickness of 2mm, size 100mm×100mm. The experimental setup is shown in Figure 2 (a), and the actuation voltage, irradiation time of HIFU and drawing speed, and position of the acrylic board were varied.

The RF power amplifier was a RAD model ZSA4242-10K250M (fixed gain of 42db). Figure 2 (b), (e) shows a graph of focal temperature at each voltage and color mapping of the dots created for each voltage. The thermal energy of HIFU is sufficient to melt the acrylic plate since the acrylic plate melts at temperatures above 90 degrees Celsius. To create dots, HIFU was irradiation for 1, 5, 10, 15, and 20 seconds and experiments were performed 5 times.

3 HANDS-ON PROCEDURE

The use of HIFU for processing acrylic sheets has made it possible to easily and quickly control transparency, thickness, embossment, and tactile feedback. In the demo, participants will be able to experience various features of HIFU processing.

Transparency and Thickness: The line thickness of the embossment can be adjusted (from 0.87 to 3mm) by changing the distance between the HIFU and the plate, and the transparency can be changed by adjusting the amplitude of HIFU actuation Figure 1 (b), (d). **Tactile:** The roundness and sharpness of the feature can be adjusted by changing the irradiation time Figure 1 (e). Round dots have potential to be used as braille in the future. **Surface Texturing:** By randomly irradiate the plate, it is possible to apply textures to the acrylic surface Figure 1 (f). **Caustics:** The ability to apply embossment without changing the transparency opens up the possibility of making caustics lens in the future. We demonstrate its potential by creating shadow arts using HIFU. While the figures in the paper is generated using XYZ plotter, the pattern can be drawn easily free hand, and we will provide opportunity to try it out by themselves.

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REFERENCES

- Mohsen Habibi, Shervin Foroughi, Vahid Karamzadeh, and Muthukumaran Packirisamy. 2022. Direct sound printing. *Nature Communications* 13, 1 (2022), 1800.
- Kai Melde, Heiner Kremer, Minghui Shi, Senne Seneca, Christoph Frey, Ilia Platzman, Christian Degel, Daniel Schmitt, Bernhard Schölkopf, and Peer Fischer. 2023. Compact holographic sound fields enable rapid one-step assembly of matter in 3D. *Science Advances* 9, 6 (2023), eadf6182.